

Total No. of Questions : 8]

SEAT No. :

[Total No. of Pages : 2

P417

[6003]-505

T.E. (E & CE)

EMBEDDED PROCESSOR & APPLICATIONS

(2019 Pattern) (Semester - II) (310354)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data if necessary.

Q1) a) Interface of LPC 2148 with PC using UART. Draw interfacing diagram and explain it. [9]

b) What are the advantages of using LPC 2148 instead of other processors in embedded system? [9]

OR

Q2) a) Write down the characteristic features of UART of LPC 2148. [4]

b) Write an embedded C program to transmit character 'A' to PC. [6]

c) Draw and explain interfacing of on chip DAC with LPC2148 with flowchart. [8]

Q3) a) What are the features and advantages of ARM CORTEX M3 in embedded system. [8]

b) Compare CORTEX A, CORTEX M, CORTEX R processors. [5]

c) Write a note on ARM processor development. [5]

OR

Q4) a) Write down the specifications of ARM CORTEX M3. [5]

b) Draw detailed architecture of ARM CORTEX M3. [8]

c) Compare between ARM Cortex-M3 and ARM-7 [5]

Q5) a) Explain programmer model of ARM CORTEX M4 with neat labelled diagram. [9]

b) Explain an algorithm to send "SPPU" serially via STM32F4xx controller to Desktop PC on Hyper Terminal. Assume (UART, 9600 Baud Rate). [8]

OR

P.T.O.

- Q6)** a) Write a note on different types of timers of STM32F4xx. [6]
b) Draw an diagram or flowchart to interface Seven segments LED with STM32F4xx microcontroller. [6]
c) Compare ARM Cortex M3 with ARM Cortex M4. [5]

- Q7)** a) What is IoT? Enlist the characteristics of IoT. [8]
b) Write a note on Sensors and actuators. [6]
c) Write down the characteristics of Embedded System. [3]

OR

- Q8)** a) Draw and explain basic block diagram of Embedded System with IoT.[8]
b) Explain any one of the below with flow/block diagram. [9]
Agriculture automation.

OR

Health monitoring system.



Total No. of Questions : 8]

SEAT No. :

PC-1796

[Total No. of Pages : 2

[6353] - 115

T.E. (Electronics & Computer Engineering)
EMBEDDED PROCESSORS AND APPLICATIONS
(2019 Pattern) (Semester - II) (310354)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q1 or Q2 , Q3 or Q4, Q5 or Q6 and Q7 or Q8.
- 2) Figures to the right side indicate full marks.
- 3) Assume suitable data, if necessary.

- Q1)** a) Draw interfacing diagram and write an embedded C program to transmit character 'Y' to PC using LPC2148. [6]
- b) Write the features of ADC of LPC2148. Draw block diagram of ADC of LPC2148. [6]
- c) Explain DAC register of LPC2148. Write an embedded C Program for generation of square waveform using on chip DAC of LPC2148 with its interfacing diagram. [8]

OR

- Q2)** a) Draw interfacing of GSM module with LPC 2148. Write algorithm and draw flowchart for sending the message through GSM module. [6]
- b) Draw the architecture of UART0 of LPC 2148 and explain the function of each block in detail. [6]
- c) Draw and explain interfacing of 12C EEPROM with LPC2148. Also write algorithm and draw flow chart for the same [8]
- Q3)** a) Write the features of Cortex M3 processor. Draw and explain the block diagram of Cortex M3 processor. [8]
- b) Draw and explain in detail CMSIS structure of cortex series. [8]

P.T.O.

OR

- Q4)** a) Write features of Cortex R series. Compare Cortex A series, Cortex R series and Cortex M series. [8]
- b) Compare ARM7 and ARM Cortex -M3 series. Also list out advantages of ARM Cortex -M3 for embedded application. [8]
- Q5)** a) Draw and explain the block diagram of ARM Cortex M4 processor. Also list out the features of same. [8]
- b) Explain with example following GPIO registers of STM32F4xx series. [8]
- i) GPIOx_MODER ii) GPIOx_IDR iii) GPIOx_ODR
- iv) GPIOx_BSSR

OR

- Q6)** a) With the help of diagram explain in detail programmers model of ARM Cortex M4 processor. [8]
- b) Draw interfacing of seven segment display to STM32F4xx series. Write a program to display 0 to 9 count on seven segment. [8]
- Q7)** a) Explain Sensors and Actuators with suitable examples. Compare Sensors and Actuators. [9]
- b) Explain case study of Smart Home Automation using IoT with detailed diagram. [9]

OR

- Q8)** a) Define Internet of Things. Draw and explain the architecture of Internet of Things. [9]
- b) Explain case study of Agriculture Automation using IoT with detailed diagram. [9]



Total No. of Questions : 8]

SEAT No. :

PB-3854

[Total No. of Pages : 2

[6262]-116

T.E. (Electronics and Computer Engg.)

EMBEDDED PROCESSORS & APPLICATIONS

(2019 Pattern) (Semester - II) (310354)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Solve Q.1 OR Q.2, Q.3 OR Q.4, Q.5 OR Q.6 and Q.7 OR Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Assume Suitable data, if necessary.

- Q1)** a) What is interfacing? What are the advantages of using LPC2148 instead of other processors in embedded system? [9]
- b) Draw and explain interfacing of on chip DAC with LPC2148 with flowchart. [9]

OR

- Q2)** a) Write down the characteristic features of UART of LPC2148. [4]
- b) Write an embedded C program to transmit character 'A' to PC. [6]
- c) Interface of LPC 2148 with PC using UART. Draw interfacing diagram and explain it. [8]

- Q3)** a) What are the features and advantages of ARM CORTEX M3 in embedded system [8]
- b) Differentiate CORTEX A, CORTEX M, CORTEX R processors. [5]
- c) Write a note on ARM processor development. [5]

OR

- Q4)** a) Write down the specifications of ARM CORTEX M3. [5]
- b) Draw detailed architecture of ARM CORTEX M3. [8]
- c) Compare between ARM Cortex-M3 and ARM 7 [5]

P.T.O.

Q5) a) Explain programmer model of ARM CORTEX M4 with neat labelled diagram. [9]

b) Desktop PC on HyperTerminal. Assume (UART, 9600 Baud Rate) [8]

OR

Q6) a) Write a note on different types of timers of STM32F4xx. [6]

b) Draw an diagram or flowchart to interface Seven segments LED with STM32F4xx microcontroller. [6]

c) Compare ARM Cortex M3 with ARM Cortex M4 [5]

Q7) a) What is IoT? Enlist the characteristics of IoT. [8]

b) Write a note on Sensors and actuators [6]

c) Write down the characteristics of Embedded System [3]

OR

Q8) a) Draw and explain basic block diagram of Embedded System with IoT [8]

b) Explain any one of the below with flow/block diagram
Agriculture automation [9]

OR

Health monitoring system



Total No. of Questions : 8]

SEAT No. :

[Total No. of Pages : 2

P7613

[6180]-132

T.E. (E&C.E)

EMBEDDED PROCESSOR & APPLICATIONS

(2019 Pattern) (Semester-II) (310354)

Time : 2½Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8.
- 2) Draw neat labeled diagrams wherever necessary
- 3) Figures to the right side indicate full marks.
- 4) Assume Suitable data if necessary.

- Q1)** a) Compare ARM Cortex M3 with ARM7 TDMI [4]
b) Write down the features of UART of LPC2148, Write an embedded C program to transmit character 'E' to PC. [6]
c) Draw and explain interfacing of EEPROM using 12C communication to LPC2148. [8]

OR

- Q2)** a) Explain any 3 instructions of ARM7. [6]
b) List the features of on chip ADC of LPC2148. [8]
c) List the features of LPC2148. [4]

- Q3)** a) Explain any 4 instructions of ARM CORTEX M3. [8]
b) Compare CORTEX A, CORTEX M, CORTEX R processors [5]
c) Write a note on ARM processor development. [5]

OR

- Q4)** a) Write down the specifications of ARM CORTEX M3. [5]
b) Draw detailed architecture of ARM CORTEX M3 [8]
c) Enlist the advantages of ARM Cortex-M3 [5]

- Q5)** a) Enlist various registers required to configure Serial Communication of STM32F4xx Microcontroller. Explain any two with suitable example [9]
b) Explain an algorithm to send "ENGINEER" serially via STM32F4xx controller to Desktop PC on HyperTerminal. Assume (UART, 9600 Baud Rate) [8]

OR

P.T.O.

- Q6)** a) Write a note on different types of registers of STM32F4xx. [7]
b) Draw an interfacing diagram to interface push button and LED using STM32F4xx microcontroller. [5]
c) Compare ARM Cortex M3 with ARM Cortex M4. [5]
- Q7)** a) Explain in brief Internet of Things? Draw its architecture. [8]
b) Write a note on Sensors and actuators. [6]
c) Write down the characteristics of Embedded System. [3]

OR

- Q8)** a) Draw and explain basic block diagram of Embedded System with IoT.[8]
b) Explain any one of the below with flow/block diagram. [9]
Smart Home automation

OR

Waste Management for Smart City using IoT

